

Credit OK: Looking beyond financial data to help micro entrepreneurs secure working capital

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ABOUT CREDIT OK



Credit OK developed a credit scoring methodology that provides virtual credit solutions deployed on Google Cloud to help businesses and SMEs secure working capital loans.

About Credit OK

Credit OK is a fintech startup in Thailand on a mission to solve financial inclusion problems in the country. It developed its own credit scoring methodology using alternative data sources such as transaction data, application filling behavior, and psychometrics to analyze risk and help small-scale businesses in the supply chain ecosystem to get loans without going through credit bureau approval.

Industries: Financial Services & Insurance

Google Cloud results

- Reduces time to build an end-to-end data analytics and ML architecture from two months to just three weeks
- Building a big data startup without hiring any infrastructure engineers
- Increases deployment speed by 10x with BigQuery

Enabling data with zero maintenance infra

Location: Thailand

Functions

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Functions (<https://cloud.google.com/functions/>)

Cloud Run (<https://cloud.google.com/run/>)

Kubernetes Engine (<https://cloud.google.com/kubernetes-engine/>)

Cloud IAM (<https://cloud.google.com/iam/>)

When CEO and co-founder of Credit OK

(<https://www.creditok.co/>), Jet Tanthanapongphan,

started the company in 2018, he had a clear

mission. It was to help Thai small businesses and

micro enterprises in the supply chain and B2B sector

get loans to fund their business. Many of these

businesses transact in cash in the initial stages, so a

lot of revenue is undocumented, making it a challenge for micro, small, and medium-sized businesses (SMBs) to qualify for loans that require credit bureau approval. This issue was close to the heart for Jet, who had experienced this problem firsthand in a previous venture. Although he was fortunate enough to fund his business from his own savings, many entrepreneurs don't have the same luxury. That's why he decided to start Credit OK, a fintech company that provides a customized credit scoring model specific to industries, to enable a more inclusive system for assessment of credit worthiness.

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Credit OK*

Founded as a cloud-native business, Credit OK was part of the Google Cloud for Startups (/developers/startups) program where it received access to technical documentation and online courses to develop the team's expertise on Google Cloud products. At the same time, the support Credit OK received from the Google Cloud for Startups program also enabled it to quickly ramp up new hires and equip them with the right training for relevant Google Cloud products. After learning more about the platform's benefits, it decided to adopt Google Cloud (<https://cloud.google.com/>), particularly for its big data and analytics capabilities.

"At a glance, other leading cloud providers provide very similar services, but the tool that led us to select Google Cloud is BigQuery (/bigquery). Because we're always working with data analytics and are specialized in SQL, BigQuery is the perfect match for

us," says Gap Chaonithi, CTO and co-founder of Credit OK.

"We really experienced a difference with BigQuery, compared to other data analytics solutions we had tried before. It has sped up the data analysis process with a 10-fold increase in credit risk model deployment speed. We don't see slowdowns in performance or issues such as bottlenecks, even when

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A secure data warehousing solution with transparent infrastructure

Before adopting Google Cloud, Credit OK had experimented with different cloud technologies, but realized that they couldn't meet the requirements it wanted in the crucial area of data ingestion and storage. When Gap's team discovered BigQuery, it knew this was the best option due to its ease of use and how it enables Credit OK to securely store all of its data in one place. "We really experienced a difference with BigQuery, compared to other data analytics solutions we had tried before. It has sped up the data analysis process with a 10-fold increase in credit risk model deployment speed. We don't see slowdowns in performance or issues such as bottlenecks, even when running lots of data," says Palm Phuwarat, Chief Product and Data Officer at Credit OK.

For a more inclusive loan qualification process, Credit OK provides an end-to-end financial solution for customers where it holds the risk between distributors and small businesses by giving them a credit line so they can purchase freely from

distributors. Credit OK looks at different types of information as well as financial data in the qualifying process, which is why having a reliable data ingestion system is key. "Data ingestion from partners is very important to us in building a risk model because we rely on it for underwriting. Google Cloud enables us to build the right features that feed into our model, so that we can ingest different types of data into the system without worrying about infrastructure," says Palm.

But perhaps the most important piece of the puzzle is the company's web application. After companies approach Credit OK through its partners' distribution location, the store manager will fill in a form with detailed business information as well as their customer ID, which is linked to a pre-scored database that Credit OK ingests from the partner. The transaction information and purchasing behavior from the partner helps Credit OK assess their potential business growth and their willingness to repay. The company is then asked to fill in a form with more detailed business information that includes any form of cash transaction, as well as psychometric elements like skills and competency, which could impact the potential of business growth.

These details enter an underwriting process, and that's where the inclusive loan qualification process begins. Credit OK uses its credit scoring module, deployed on [Cloud Functions](#) (/functions), to process customer data and compute a credit score.

Because this web application is built on Google Cloud, the team builds CI/CD pipelines that automate the application's quality assurance testing and deploy them on Cloud Run, before serving them to partners. With so much movement of traffic and data taking place at any one time, Gap says that the autoscaling feature of Cloud Run has been beneficial as it takes away the need for infrastructure management. Now the team only needs three weeks to build a full data pipeline that includes data ingestion, data cleaning, data exploration, building a model, and deploying it. This is a big time saver compared to two months, which is the amount of time it would otherwise take.

For the more complex tasks such as running special data pipeline tasks in background or serving web applications on specific IP addresses, the team relies on Google Kubernetes Engine (/kubernetes-engine) (GKE). On top of that, the flexibility of GKE enables Credit OK to deploy other open source platforms on GKE with little customization when required.



Running large-scale operations with a lean team

As a startup, Credit OK is excited by the ability to run big data and machine learning (ML) operations with a lean team, while Google Cloud takes care of its own infrastructure maintenance. "I've tried building an infrastructure on my own, and it's really frustrating to create a virtual machine, install everything, then find that it's always down and requires constant maintenance," Gap says.

Usually, a big data application like Credit OK would require at least one employee dedicated to setting up and managing it. But with Google Cloud, the company only needs to enter the web console, import data, deploy the application, and set up the CI/CD pipeline. Being able to hand off infrastructure management saves Credit OK the cost of hiring additional DevOps engineers, which it is then able to reinvest in other areas of the business.

Meanwhile, Gap manages security of Credit OK's application using [Cloud Identity and Access Management \(/iam\)](#), a part of the [Google Cloud Architecture Framework \(/architecture/framework\)](#), which enables full control and visibility of all Google Cloud resources within the organization. "If an employee needs access to a specific resource, I send them the relevant documents to read up on, then give access so they can start using it immediately. Google Cloud products are pretty intuitive, so no real training is required."

According to Gap, the assurance that its web application and data warehouse will not fail is most valuable to the company. "Having a stable platform supported by Google Cloud allows us to focus on improving our processes and credit scoring models by consistently exploring different ML capabilities."

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—Gap Chaonithi, CTO and co-founder, Credit OK

Helping businesses to raise capital during the pandemic

At the outset of COVID-19, many micro SMEs were heavily impacted, and Credit OK saw a rising demand for working capital credit. The company wanted to do its part to support small businesses through this challenging season.

Because Credit OK's customers are mainly in the construction and supply chain sector, everyone it lends to was impacted. Credit OK decided to adjust its loan repayment criteria to support them, as this pandemic was a unique situation that had never happened before. To address this issue, the company introduced three different loan restructuring programs for customers depending on the severity of COVID-19 impact to their business.

From reducing installments and extending repayment periods to offering loan payment deferments, Credit OK had to adjust the structure of its database significantly. "With Google Cloud, we were able to quickly roll out new repayment features to our customers during the COVID-19 pandemic in under a month with the help of Cloud Run, without any hiccups in the system."



Bigger plans in the pipeline with machine learning

The Credit OK team is fascinated with the potential of big data and ML to continually improve the company's service offerings. It believes that the key to enhancing the credit scoring process further is to explore image recognition and optical character recognition (OCR) technologies. Having these capabilities would enable the company to quickly scan receipts from cash transactions, for example, and other relevant documents that would support a business' loan application.

"We keep learning and finding ways to improve how we use Google Cloud solutions," says Gap. "Now that we have room to further explore the area of ML, we're looking forward to building credit scoring models better and faster, so that more businesses can have the opportunity to thrive."